12. How many ways a committee of six persons from 10 persons can be chosen along with a chair person and a secretary?
13. How many different selections of 5 books can be made from 12 different books if,
    (i) Two particular books are always selected?
    (ii) Two particular books are never selected?
14. There are 5 teachers and 20 students. Out of them a committee of 2 teachers and 3 students is to be formed. Find the number of ways in which this can be done. Further find in how many of these committees
    (i) a particular teacher is included?
    (ii) a particular student is excluded?
15. In an examination a student has to answer 5 questions, out of 9 questions in which 2 are compulsory. In how many ways a student can answer the questions?
16. Determine the number of 5 card combinations out of a deck of 52 cards if there is exactly three aces in each combination.
17. Find the number of ways of forming a committee of 5 members out of 7 Indians and 5 Americans, so that always Indians will be the majority in the committee.
18. A committee of 7 peoples has to be formed from 8 men and 4 women. In how many ways can this be done when the committee consists of
    (i) exactly 3 women?
    (ii) at least 3 women?
    (iii) at most 3 women?
19. 7 relatives of a man comprises 4 ladies and 3 gentlemen, his wife also has 7 relatives; 3 of them are ladies and 4 gentlemen. In how many ways can they invite a dinner party of 3 ladies and 3 gentlemen so that there are 3 of man's relative and 3 of the wife' s relatives?
20. A box contains two white balls, three black balls and four red balls. In how many ways can three balls be drawn from the box, if at least one black ball is to be included in the draw?
21. Find the number of strings of 4 letters that can be formed with the letters of the word EXAMINATION?.
22. How many triangles can be formed by joining 15 points on the plane, in which no line joining any three points?
23. How many triangles can be formed by 15 points, in which 7 of them lie on one line and the remaining 8 on another parallel line?
24. There are 11 points in a plane. No three of these lies in the same straight line except 4 points, which are collinear. Find,
    (i) the number of straight lines that can be obtained from the pairs of these points?
    (ii) the number of triangles that can be formed for which the points are their vertices?
25. A polygon has 90 diagonals. Find the number of its sides?

4.6 Mathematical induction

Let us consider the sum of the first n positive odd numbers. These are $1, 3, 5, 7, \cdots, 2n-1$. The first odd number 1 which is equal to 1. The first two odd numbers are 1 and 3 and their sum is 4. Writing these as follows helps us to see a pattern.

$$\begin{aligned} 1 &= 1 \\ 1+3 &= 4 \\ 1+3+5 &= 9 \\ 1+3+5+7 &= 16 \\ 1+3+5+7+9 &= 25 \end{aligned}$$